

Spacecraft Computers, Software and Data

Lab Project Overview & Team Allocation

1. Project Overview

Goal

Design and develop the platform of a 1U CubeSat (no payload), including a functional OBC, FSW, CDH, TT&C, EPS, and 1U structure. The CubeSat shall detect and switch between mission phases using FSM logic or an RTOS.

Mission context

The CubeSat flies on a stratospheric balloon. The OBC must autonomously detect and transition through six flight phases using data from an IMU and a GPS sensor.

Standby

Launch

Ascent

Cruise

Descent

Landing

Phase detection uses IMU (acceleration, rotation) and GPS (altitude, position). Data is stored onboard and transmitted to a ground station in real time.

2. Report Deliverables

#	Section	Contents
1	Introduction	Goal, objectives, general system description
2	Project Management	Gantt chart, Bill of Materials, budget
3	Systems Engineering	Requirements list, risk matrix, verification plan
4	OBC	Description, requirements, components, config, block diagram
5	FSW	Description, requirements, UML diagram, algorithmic complexity, code repository
6	CDH	Description, requirements, components, bus config, protocols, data budget
7	TT&C	Description, requirements, components, protocols, link budget
8	EPS	Power budget, battery selection, regulation, power distribution diagram
9	Verification	Test campaign description and results
10	Validation	Balloon flight opportunity: results vs mission objectives
11	Conclusions	Lessons learned, what worked, what you would do differently
App.	Appendix	Datasheets, code listings, raw data, extended calculations

3. Subsystem Allocation

Five subsystem leads, one per subsystem and one secondary for redundancy. Tasks and report deliverables listed for each.

OBC <i>On-Board Computer</i>	Team tasks • Choose and set up the microcontroller board (STM32 Nucleo-64 or equivalent) • Set up the development environment (STM32CubeIDE) for the whole team • Own the physical board - central integration point for all subsystems • Confirm hardware constraints and component availability with professor Report deliverables • Description of the OBC and its role • Subsystem requirements • Component list with justification • Board configuration (clock, pins, peripherals) • Block diagram showing OBC and connected subsystems
FSW <i>Flight Software</i>	Team tasks • Design and implement the finite state machine (6 flight phases) • Write phase detection logic using IMU and GPS data from CDH • Implement watchdog timer and safeguard memory (phase persistence on reboot) • Maintain the team code repository (Git) Report deliverables • Description of the flight software architecture • Subsystem requirements • UML diagram of the full state machine • Algorithmic complexity analysis and OBC sizing calculation • Link to code repository

CDH
*Command &
Data Handling*

Team tasks

- Wire and configure IMU (MPU-6050) and GPS (u-blox NEO-M8N) to the OBC
- Optionally add barometric pressure sensor (BMP388) for phase redundancy
- Set up I2C and SPI communication between sensors and OBC
- Implement SD card data logging - define rate, format, and file structure

Report deliverables

- Description of the CDH architecture
- Subsystem requirements
- Component list (sensors, storage)
- Bus configuration and wiring diagram
- Protocol descriptions (I2C, SPI - how each sensor communicates)
- Data budget - logging rate, storage sizing, memory allocation

TT&C
*Telemetry,
Tracking &
Command*

Team tasks

- Set up LoRa radio module (RFM95W or equivalent) on the CubeSat
- Build or configure the matching ground station receiver
- Define telemetry packet structure - what data is transmitted and how often
- Conduct ground-level radio range tests before the balloon flight

Report deliverables

- Description of the TT&C system
- Subsystem requirements
- Component list (transceiver module, antenna, ground station hardware)
- Protocol description (CCSDS or custom packet format)
- Link budget calculation - proving signal reaches ground from 30km altitude

EPS

Electrical Power System

Team tasks

- Calculate power budget - current draw per component, total consumption
- Select battery type and capacity sufficient for the full flight duration
- Choose voltage regulators to supply correct voltage to every subsystem
- Measure actual current consumption of each component during testing

Report deliverables

- Description of the power system
- Subsystem requirements
- Power budget table - all components, voltages, and current draw
- Battery selection with justification
- Power distribution diagram
- Regulation approach (LDO, DC-DC converter selection)

Name	Skills	Interested in	Allocated Task	
			Primary	Secondary
Keval	Systems Engineering, Learn FSW, Hardware (Sensors connection)	Systems Engineering, FSW	1) FSW	Systems Engineer
Moon		Everything, maybe TTC, OBC, CDH	1) TTC, 2) (Structure)	
Andrey	Hardware	FSW	1) FSW	1) TTC, 2) OBC, EPS -- review only
Frederik	Hardware (Sensors), Programming	FSW, anything maybe not EPS	1) CDH	1) FSW 2) (Structure)
Konstantin	Software, Communications	Anything besides communication lol, i.e. I would prefer EPS, FSW, OBC	1) OBC, 2) EPS, 3) secondary payload if it exists	1) CDH

4. Systems Engineer Role

One team member takes this role alongside their subsystem. Responsibilities span the full project; from requirements at the start to verification at the end.

Requirements • Write top-level mission requirements before subsystem work begins • Each subsystem lead derives their own requirements from the top-level ones • Review all subsystem requirements for completeness and correctness • Maintain requirements traceability throughout the project	Verification & validation • Design the test plan — what tests prove each requirement is met • Coordinate the test campaign across all subsystems • Document test results in the verification section of the report • Own the validation section: balloon flight results vs objectives
Project management • Maintain the Gantt chart and track milestone progress • Own the Bill of Materials and budget tracking • Build and update the risk matrix throughout the semester • Set integration milestones: plan at least 2 weeks for full-system testing	Integration coordination • Define interface agreements between subsystems early • Ensure CDH data format matches what FSW expects • Coordinate subsystem reviews at key project milestones • Flag blockers and dependencies before they become problems

5. Key Notes

Ownership vs Isolation

Each person owns their subsystem, but no subsystem is fully isolated. The integration phase is a full team effort.

Integration Timeline

Plan at least 2 weeks of full-system integration and testing before the balloon flight. Individual subsystems working in isolation is not the same as all of them working together reliably.

Questions to clarify with professor before starting

- Which microcontroller board is required: Arduino, STM32 Nucleo64, or free choice?
- What is the budget per team, and which components does the lab provide?
- Can a power bank be used as the power source, or is a custom EPS required?

First team meeting agenda

- Confirm roles
- Confirm hardware and budget situation with Ramon first
- Set up a shared code repository (GitHub) and shared document folder (probably sharepoint).
- Agree on a weekly check-in cadence (once started working on a project).
- Draft a first version of the Gantt chart together (Soon....)

Questions for Ramon:

1. Would ~200€ Payload be within the budget?
2. Do we have a LoRa receiver available or do we need to buy a new one?